

Measurement Uncertainty: ISO/TS 20914:2019 document

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Definition

Measurement uncertainty, MU:

- Parameter characterizing the dispersion of the quantity values being attributed to a measurand, based on the information used.

VIM – 3rd Ed. GCGM 2012

It describes the interval within which the value of the measurand is assumed to lie with a stated level of confidence

Why MU?

- To ensure that measurement results are **useful** and **safe** in medical practice:

- Meaningful comparison with medical **decision limits**

Screening
Diagnosis
Classification

- Meaningful comparison with **previous results**

Monitoring

➤ **ISO 15189:2012,**

5.5.1.4: “...shall determine measurement uncertainty for each measurement procedure in the **examination phase** used”

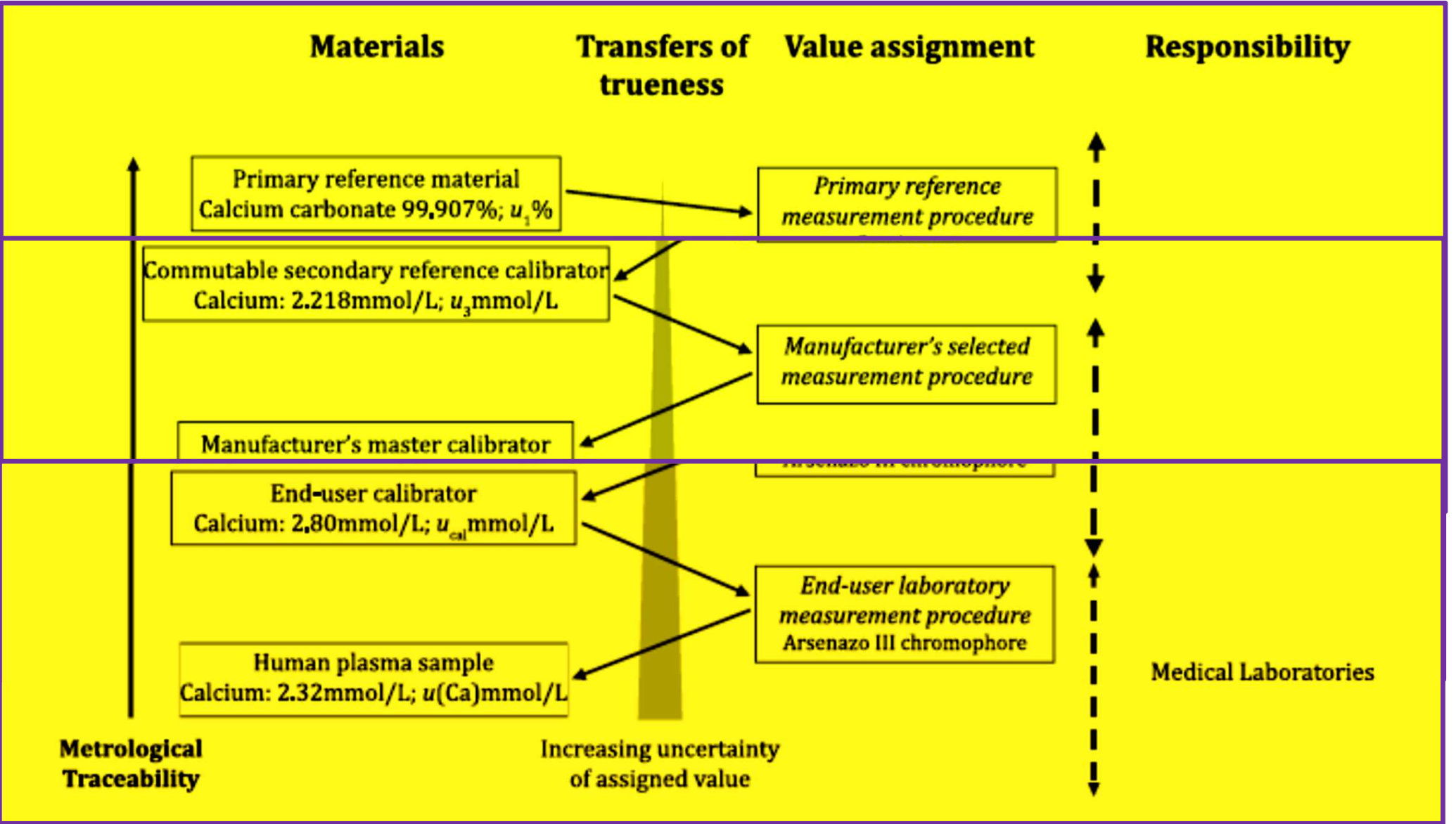
What's the Problem?

“Although all medical laboratories seeking ISO 15189:2012 accreditation know that MU estimate is a specific requirement, **few know what to do with the calculated MU.**”

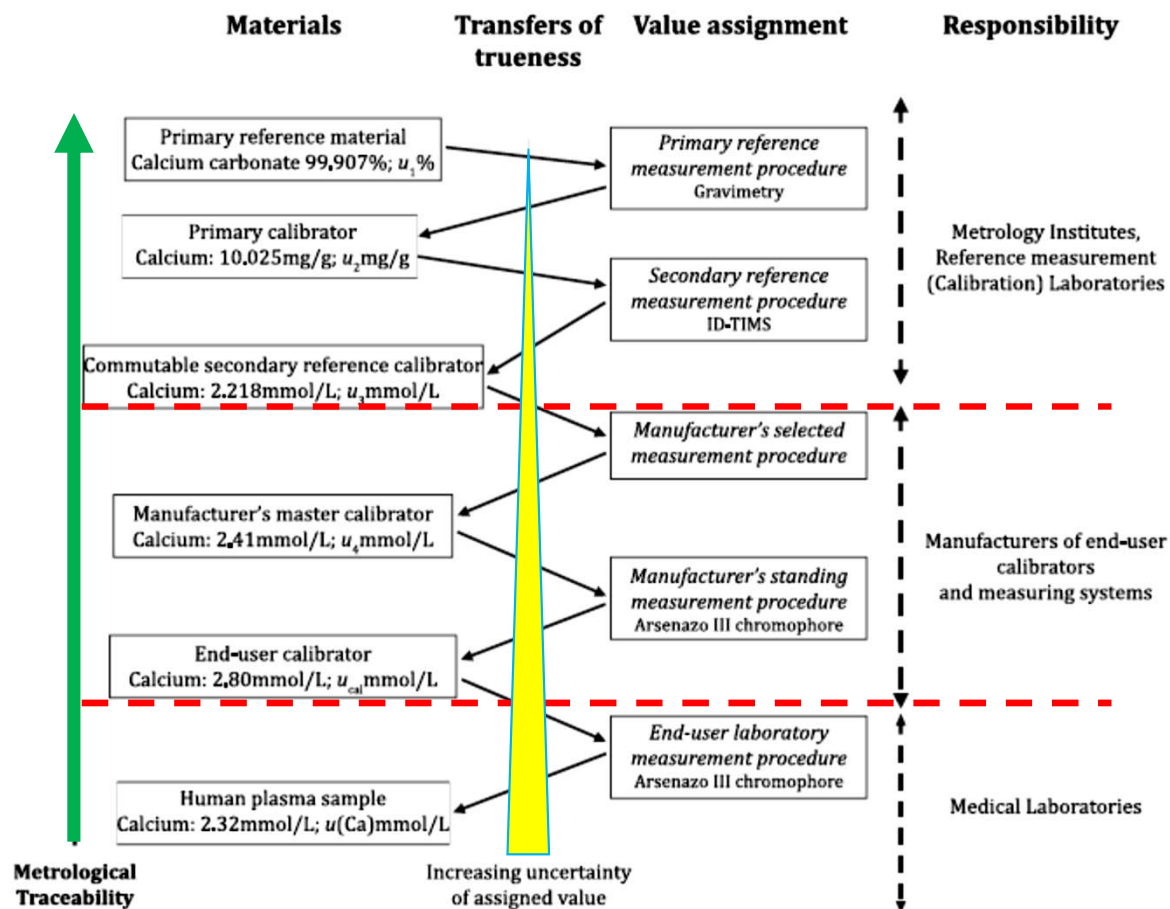
Braga F, Panteghini M. The utility of measurement uncertainty in medical laboratories. Clin Chem Lab Med 2020

“We wish to add that “**few are truly aware of how to calculate** MU in medical laboratories”.

Mario Plebani, Andrea Padoan and Laura Sciacovelli Measurement uncertainty: light in the shadows <https://doi.org/10.1515/cclm-2020-0134>



MU is all about TRACEABILITY



Example of a calibration hierarchy for a measurand with full metrological traceability to the SI

Approaches to estimating uncertainty

- **Bottom-up:**

- Determining different sources of uncertainty
- Establish a mathematical
- Estimating partial uncertainties
- Combining under the established model

- **Top-down:**

Experimental evaluation of uncertainty in the final outcome



Example

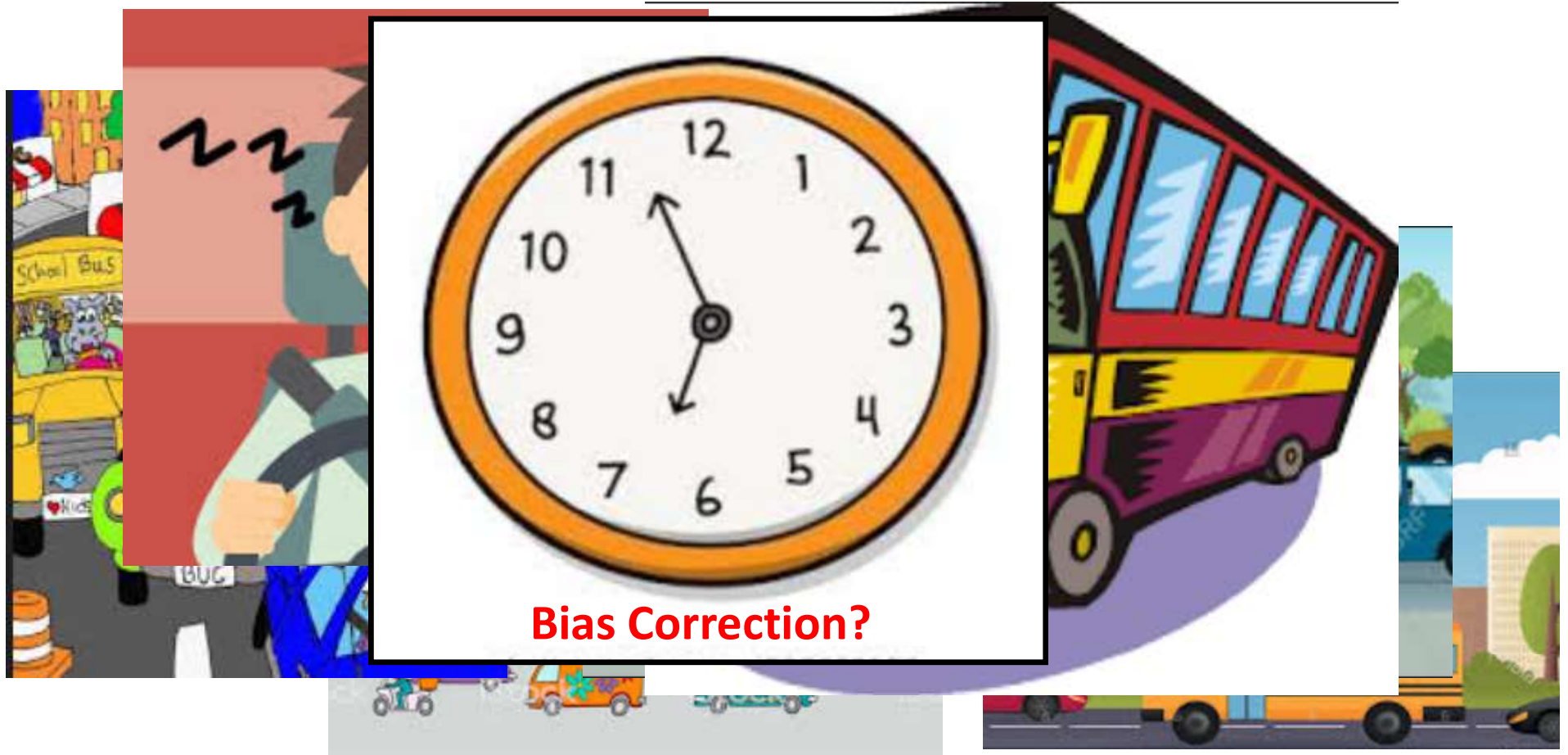
Uncertainty

Driving time:

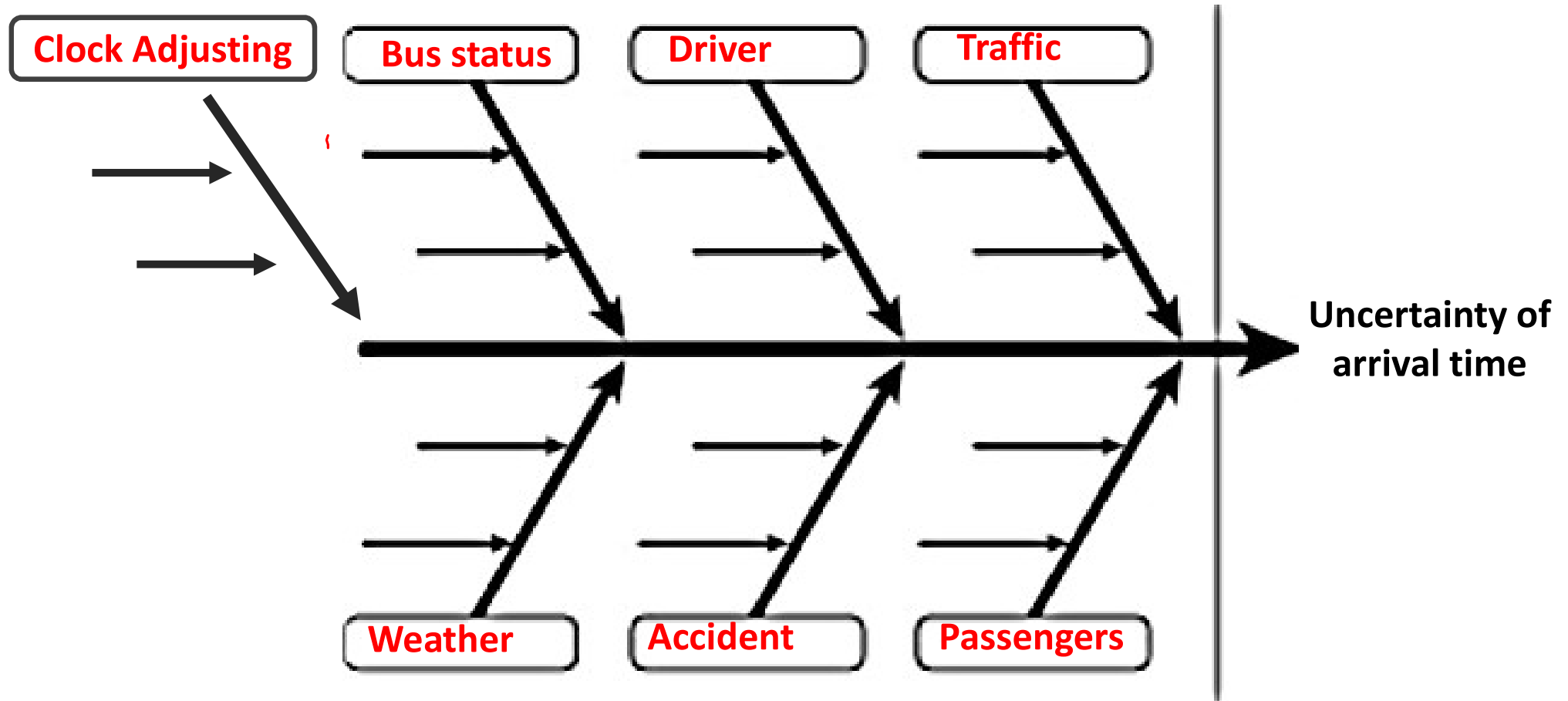
3 hrs

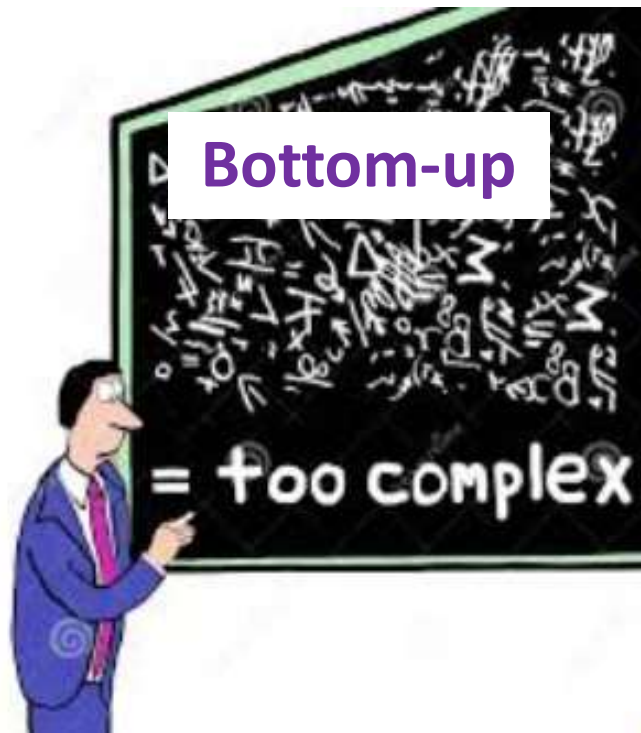


Example

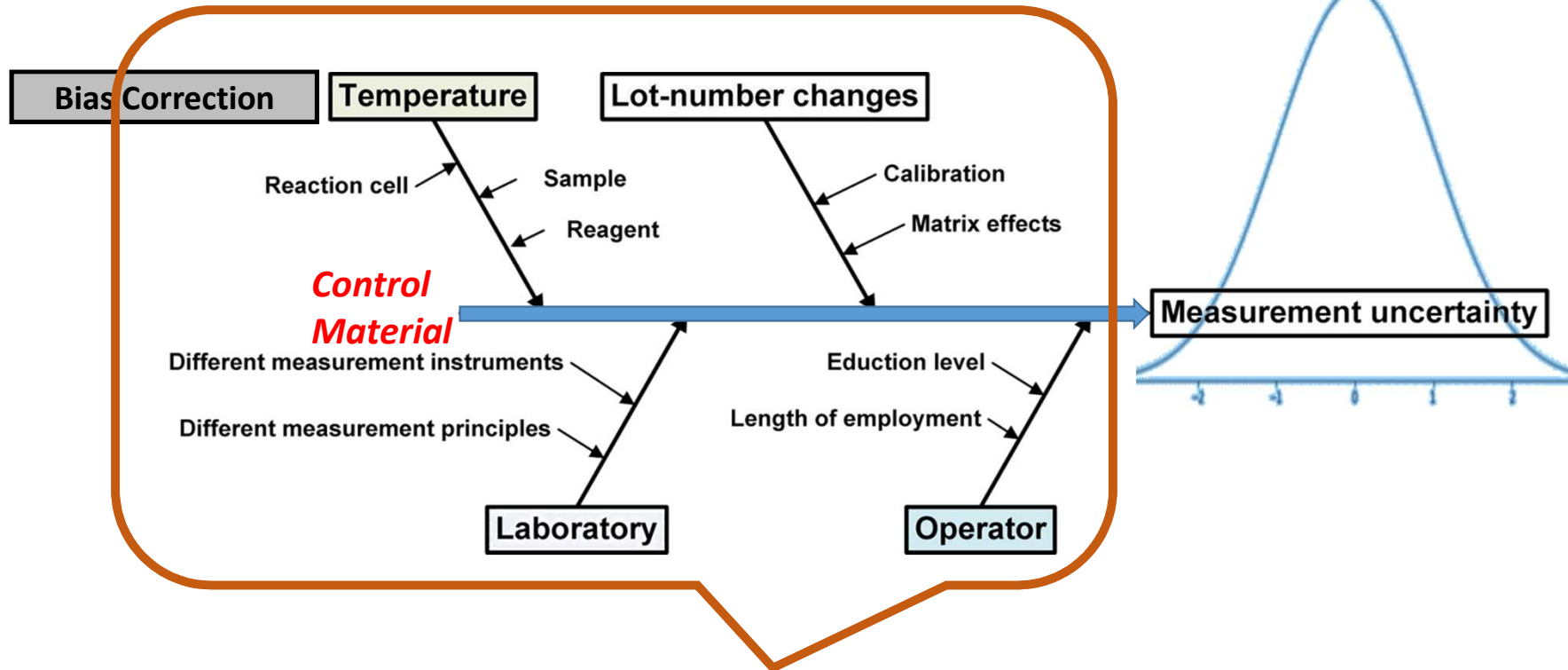


Fishbone Diagram





Sources of uncertainty in Medical Laboratory



Question: *Bottom-up* or *Top-down*?

- **Bottom-up:**

- Usually employed by **reference laboratories** to obtain ISO 17025 and 15195 certificate
- The application in medical laboratories is **too complicated** and has encountered many practical problems and objections.

- **Top-down:**

- Simpler and represents a **good alternative** to the previous approach
- Estimates MU by using **IQC, commercial calibrator information**, and **bias correction**

The utility of measurement uncertainty in medical laboratories. Braga et al. CCLM 2020; aop.
<https://doi.org/10.1515/cclm-2019-1336>

TECHNICAL
SPECIFICATION

ISO/TS
20914

First edition
2019-07

**Medical laboratories — Practical
guidance for the estimation of
measurement uncertainty**

*Laboratoires médicaux — Lignes directrices pratiques pour
l'estimation de l'incertitude de mesure*

ISO/TS 20914:2019(E)

In medical laboratory:

- The measurands are typically present in **complex biological** fluid and tissue matrices
- Characterization of commercial devices and kits is **limited** to the gathering of empirical performance data using **QC** sample
- Medical laboratory measurement procedures are well-suited to utilizing IQC and other available data to estimate MU **without the need for measurement models and complex statistics**

*“...not intended ... an exhaustive accounting of all possible uncertainties, but ... **major foreseeable sources**...”*

*“...to provide practical estimates of MU that help ensure that patient results are **fit for medical use**.”*

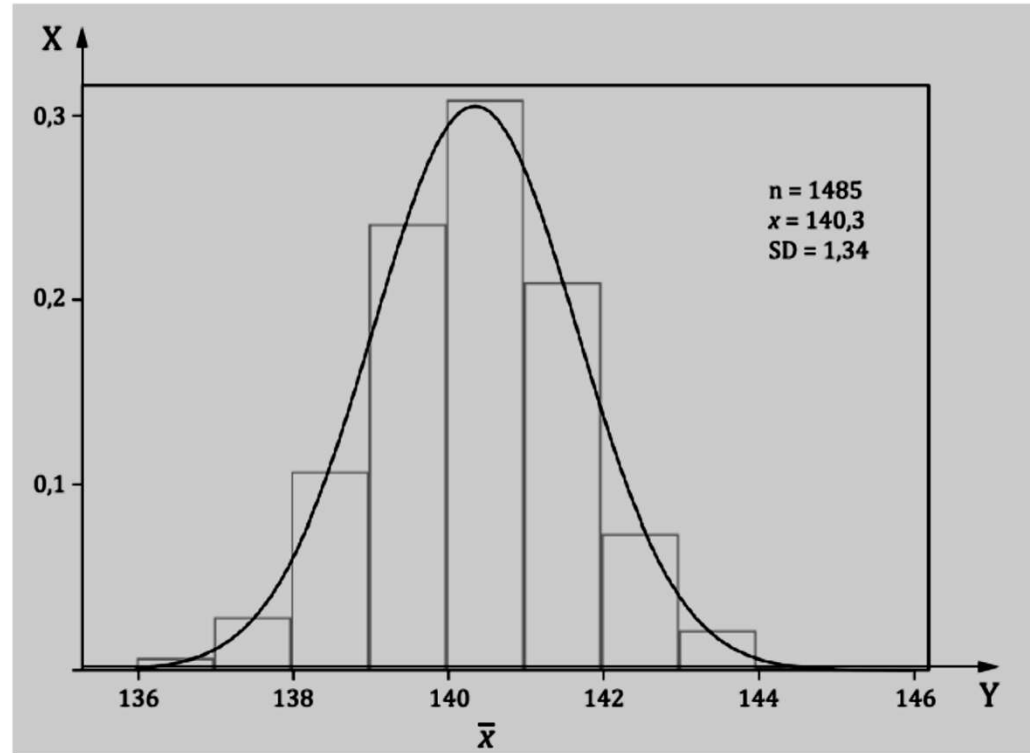
The most significant uncertainty contributions

1. Long-term imprecision data (**SD or CV**) obtained for IQC

Uncertainty of within-lab reproducibility, u_{Rw}

- *Inhomogeneity*
- *Recalibrations*
- *Reagent/Calibrator lot changes*
- *Instrumentation (e.g. electro-mechanical, maintenance, part replacement)*
- *Environment*
- *Operators*
- *Formula (e.g. approximation, assumptions, rounding)*
- *More than one system*
- *More than one measurement procedures*

Example:
Na IQC results



- An SD from such long-term precision data is termed a **standard uncertainty (u)**
- If a period sufficient to include all routine changes, the u is designated u_{RW} or the standard uncertainty under **within-laboratory conditions**

The most significant uncertainty contributions

1. Long-term imprecision (u_{Rw})

2. Uncertainty of end

3. *Occasionally* a mea

Unfortunately, the information regarding uncertainty associated with commercial calibrators is currently **very poor** when consulting calibrator package inserts.

In the majority of cases, long-term imprecision is the largest contributor



$$u(y) = \sqrt{u_{Rw}^2 + u_{cal}^2 + u_{bias}^2}$$

BIAS = Lab Mean - Reference Value

➤ *Bias correction is always imperfect!*

- Uncertainty of Reference; u_{ref}
- Uncertainty of Mean; SEM or SD_{mean}

$$SD_{\text{mean}} = SD / \sqrt{n}$$

$$u_{\text{bias}} = \sqrt{u_{\text{ref}}^2 + SD_{\text{mean}}^2}$$

Caveats about correcting bias



➤ **Only category I/II EQA are usable to evaluate bias**

- I: **Commutable**, Value assigned with **RMP or CRM**, Replicate samples
- II: **Commutable**, Value assigned with **RMP or CRM**

➤ **Correction factor not permitted in some countries because:**

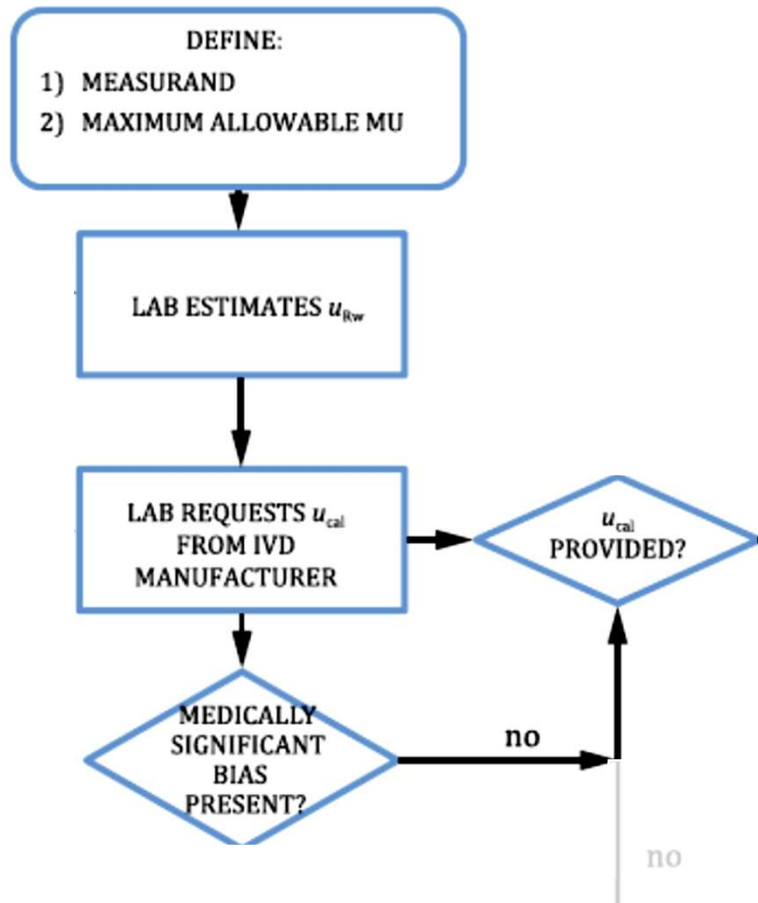
- May alter the status of the system, removing any responsibility from the manufacturer
- Depriving the system and the results of the CE marketing certification

➤ **May labs continue to use the correction factor even when the bias has been corrected in the reagent production stage**

“Therefore, we are not supporting the individual use of bias correction factors in daily practice”

The utility of measurement uncertainty in medical laboratories. Braga et al. CCLM 2020; aop.
<https://doi.org/10.1515/cclm-2019-1336>

Flowchart Diagram for calculating MU



Re-estimation of measurement uncertainty

- Apart from updating *interim estimates*, there is no need to re-estimate MU if IQC and EQA performance remain within specifications.
- Re-estimation of MU should be undertaken if there has been a *significant change* in a measuring system or a *new measurement procedure* is introduced.

POINTS
POINTS

1. More than one calibrator lot

$$u(y) = \sqrt{u_{\text{Rw}}^2 + u_{\text{cal}}^2 + u_{\text{bias}}^2}$$

- If more than one calibrator lot (e.g. 3), then **which u_{cal} ?**

$u_{\text{cal}}1$

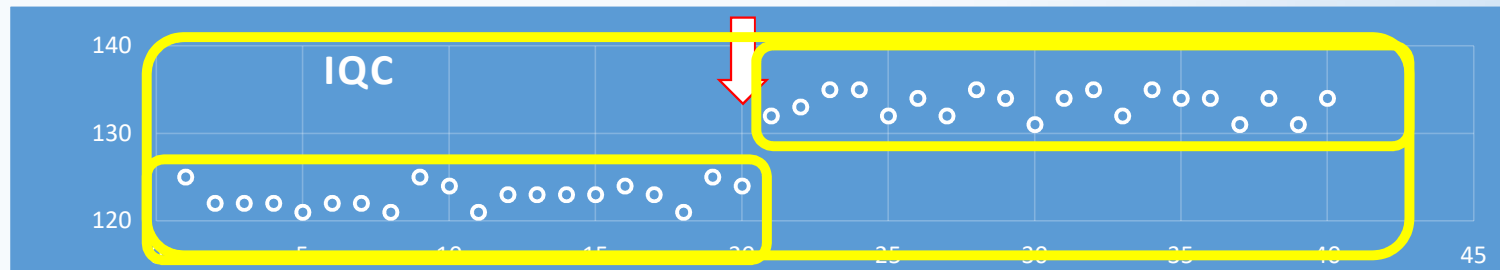
$u_{\text{cal}}2$

$u_{\text{cal}}3$

- Maybe averaging; SRMS

2. Reagent lot change

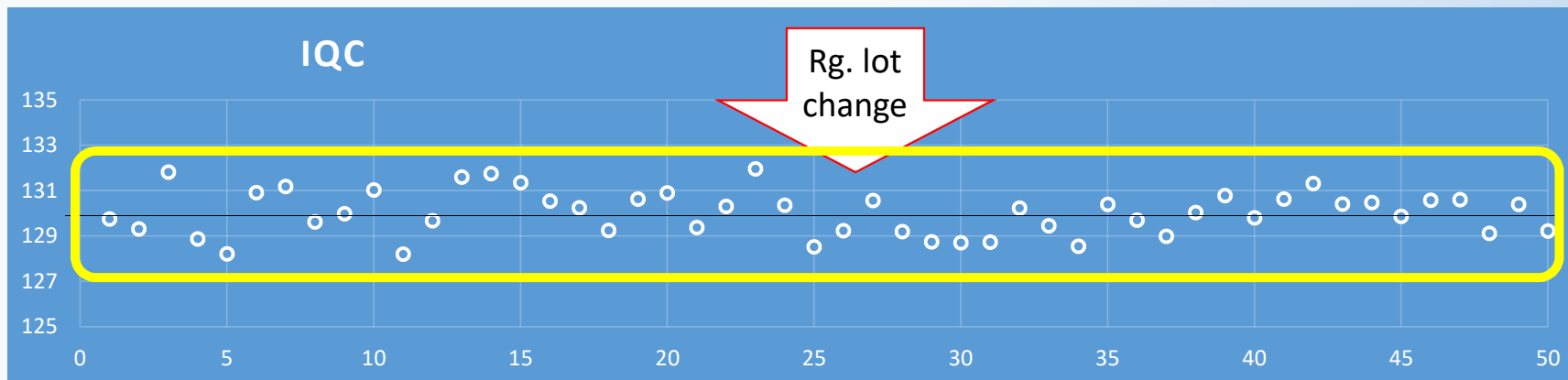
- Shift in IQC AND patient results —————> Treat IQC data as a **single set**
- Only IQC, NOT patient results —————> Treat IQC data as **separate sets**



- Patient result are shifted; but **IQC ISN'T**
- IQC Underestimates u_{Rw}

Possible Solution:

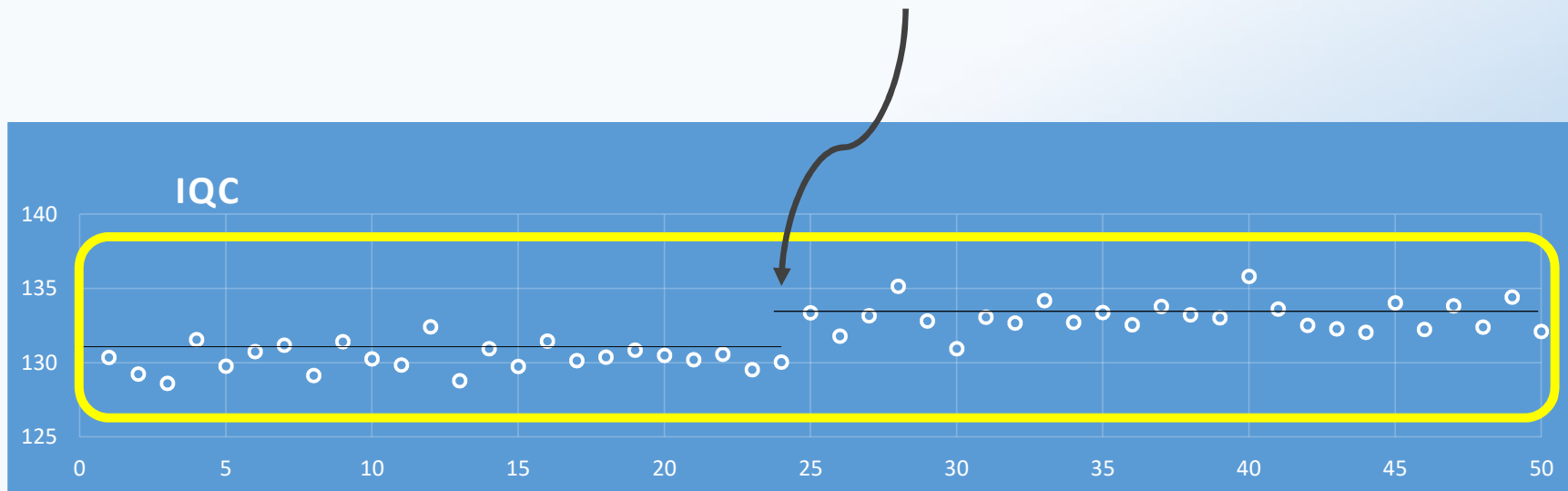
- A typical uncertainty for this “hidden shift” be estimated
- Or, IQC results be accordingly shifted



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Possible Solution:

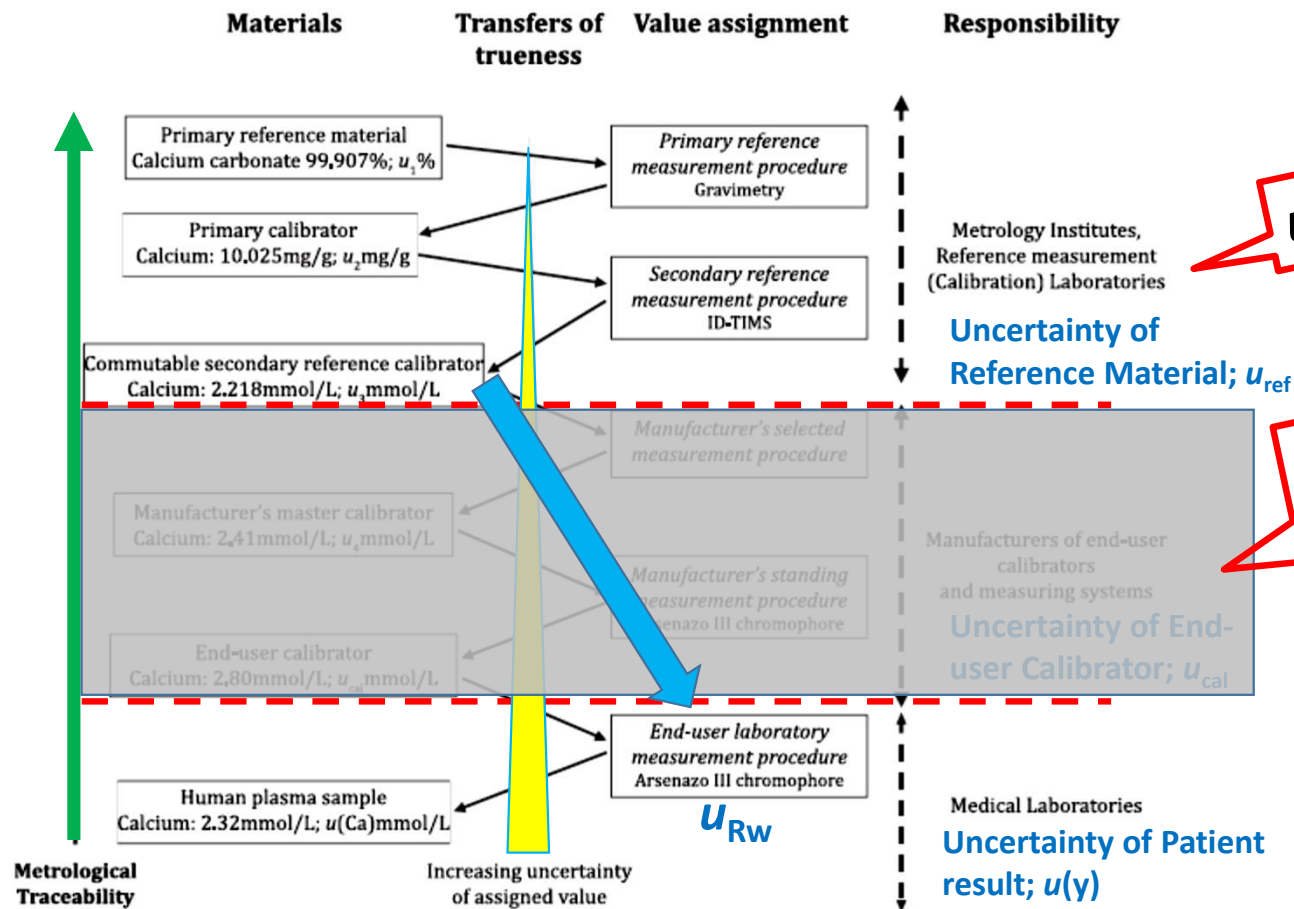
- A typical uncertainty for this “hidden shift” be estimated
- Or, IQC results be accordingly shifted



3. Bias correction

- Why yet incorporating u_{cal} ?

$$u(y) = \sqrt{u_{\text{RW}}^2 + u_{\text{cal}}^2 + u_{\text{bias}}^2}$$



u_{ref} incorporated twice!

Excessive uncertainty incorporated

If bias correction; incorporating u_{cal} :

Falsely larger uncertainty

Missing some bias as “insignificant bias”

$$u(y) = \sqrt{u_{\text{Rw}}^2 + u_{\text{cal}}^2 + u_{\text{bias}}^2}$$

$$u(y) = \sqrt{u_{\text{Rw}}^2 + u_{\text{bias}}^2}$$

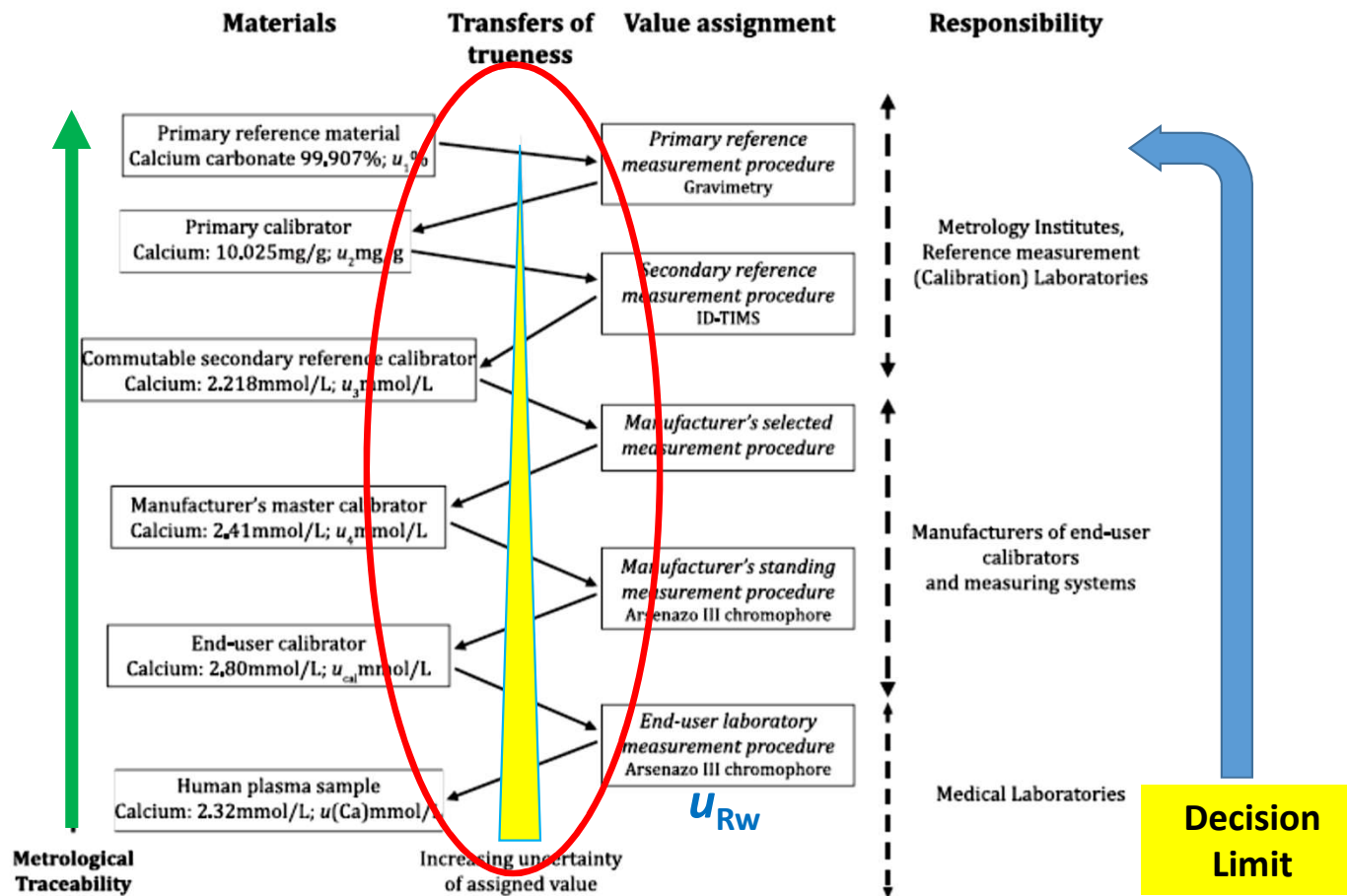
3. Monitoring & In-house RIs

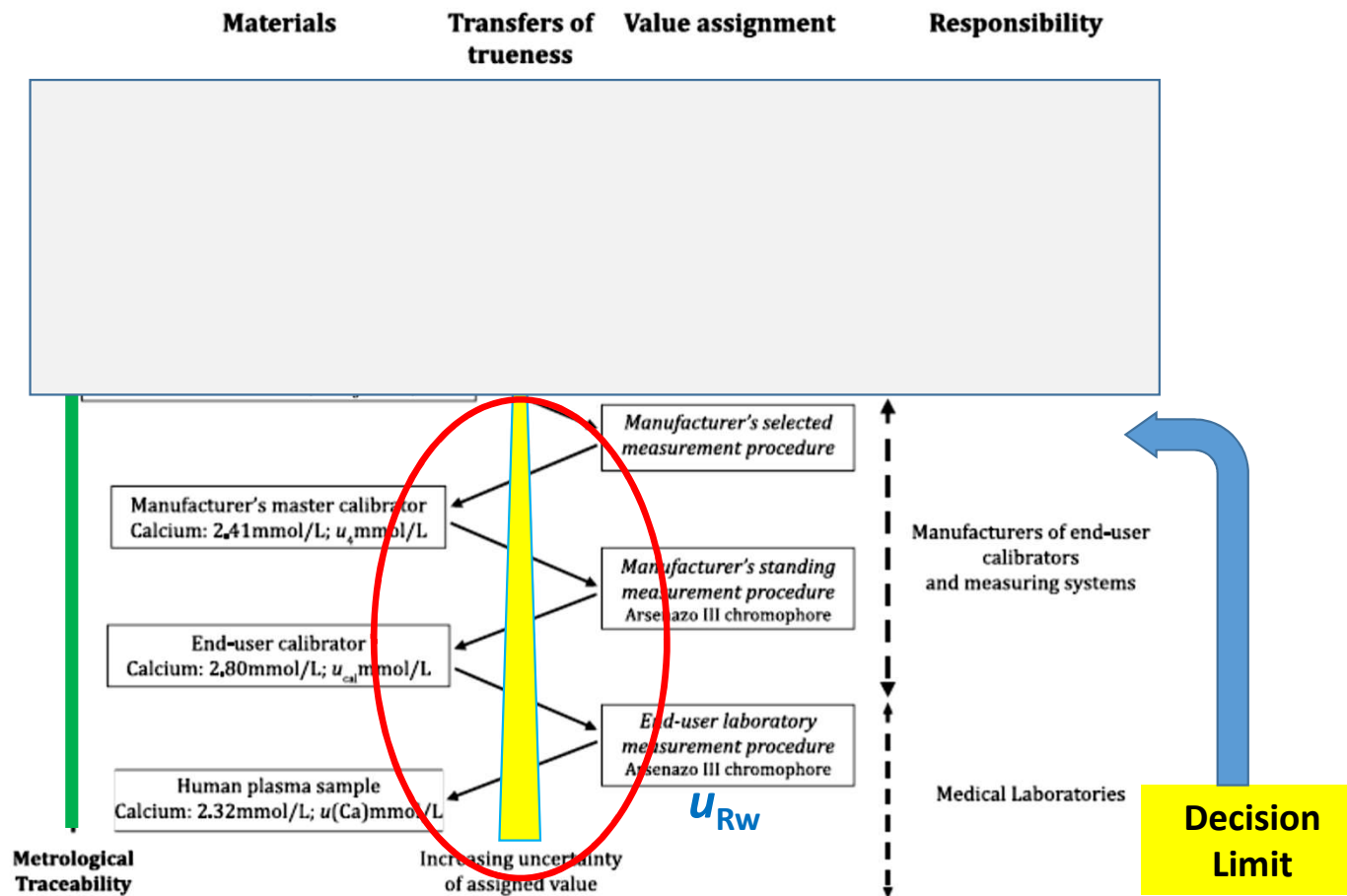
Bias is not the case when:

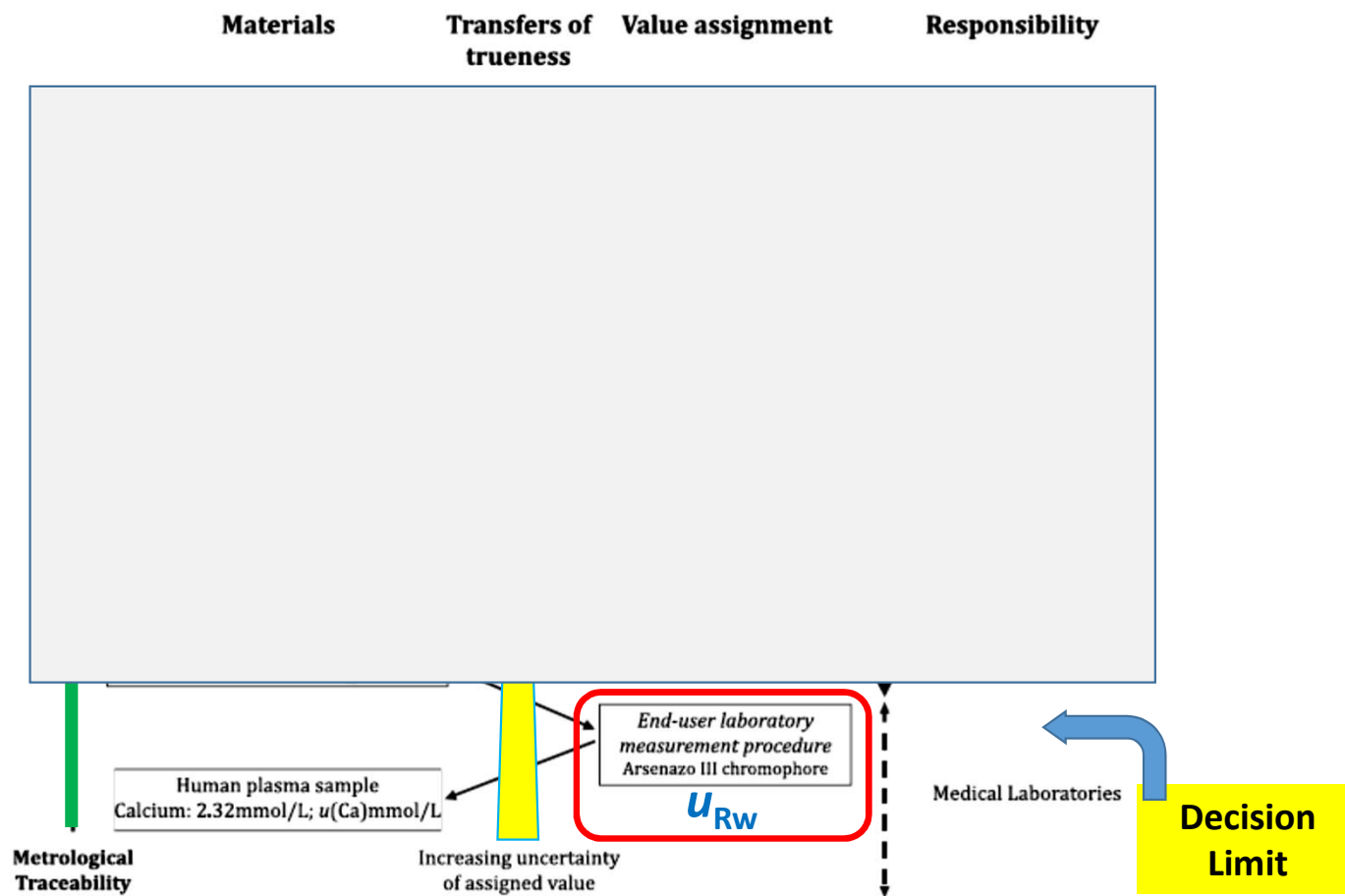
- Monitor changes in an individual over time
- In-house determined reference intervals

u_{cal} is also not the case!

Uncertainty is about traceability; so it must go up to the level the “COMPARATOR” does

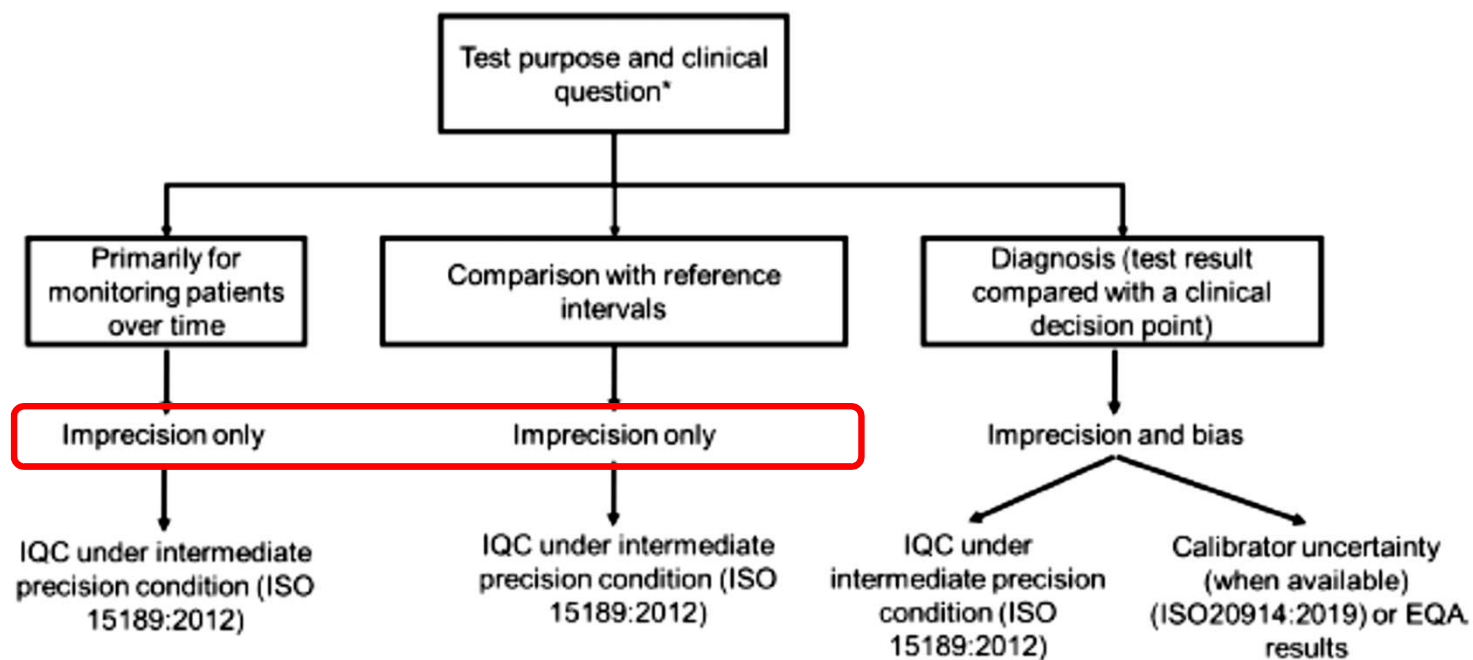






3. Monitoring & In-house RIs

2 — Plebani et al.: Measurement uncertainty: light in the shadows

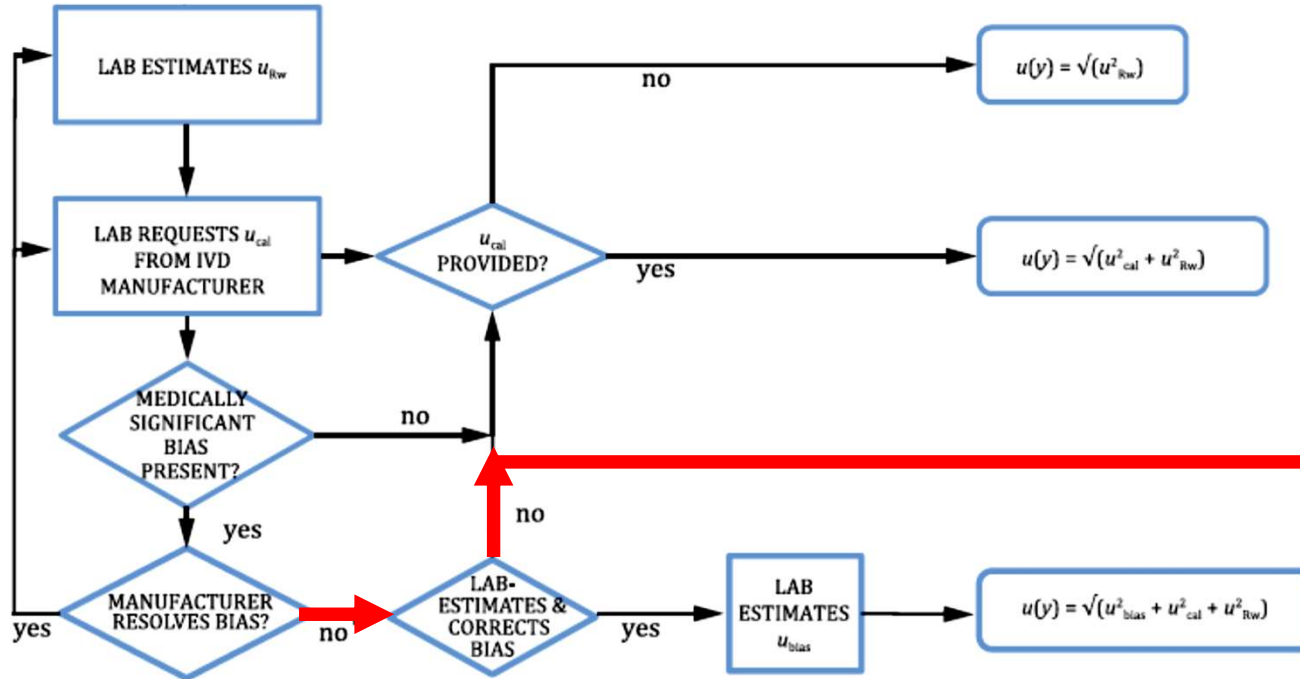


*For test results used for both monitoring and diagnosis the specific clinical question should be considered

4. Uncorrected Bias?

J. O. Westgard:

4th option: *IGNORANCE!*

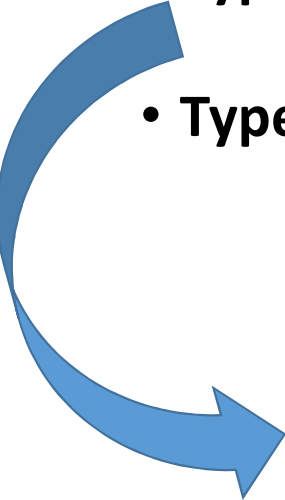




Be HAPPY!

Data sources

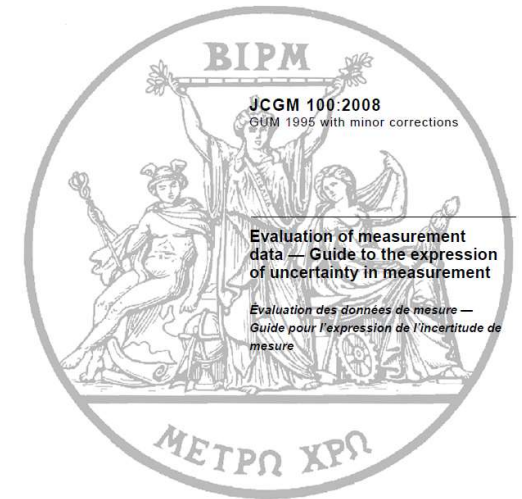
- **Type A:** Evaluation of MU from the **frequency distribution**
- **Type B:** Evaluation of MU based on experience or other information



***Standard Deviation
&
Gaussian (Normal) distribution***

Evaluation of measurement data — Guide to the expression of uncertainty in measurement (GUM) JCGM 100:2008

- 1.2: “This document is primarily concerned with the expression of uncertainty in the measurement of a **well-defined physical quantity** that can be defined by an essentially unique value.”



First edition, September 2008

*The search for any potential source of analytical uncertainty according to the “bottom-up” approach is seen as **a complicated and useless challenge***

Measurement uncertainty: light in the shadows. Plebani et al. CCLM 2020; aop. <https://doi.org/10.1515/cclm-2020-0134>

1. Uncertainty under within-lab condition, u_{cal}

In the majority of cases, **the largest contributor**

- Preferably by a **third-party** controls (i.e. different from that used to check the alignment of the measuring system)
- Controls closely resembles authentic clinical samples (ideally a **commutable** material)
- If IQC material does not undergo preparation step, a typical uncertainty for this step must be estimated

-

2. Uncertainty of end-user calibrator values, u_{cal}

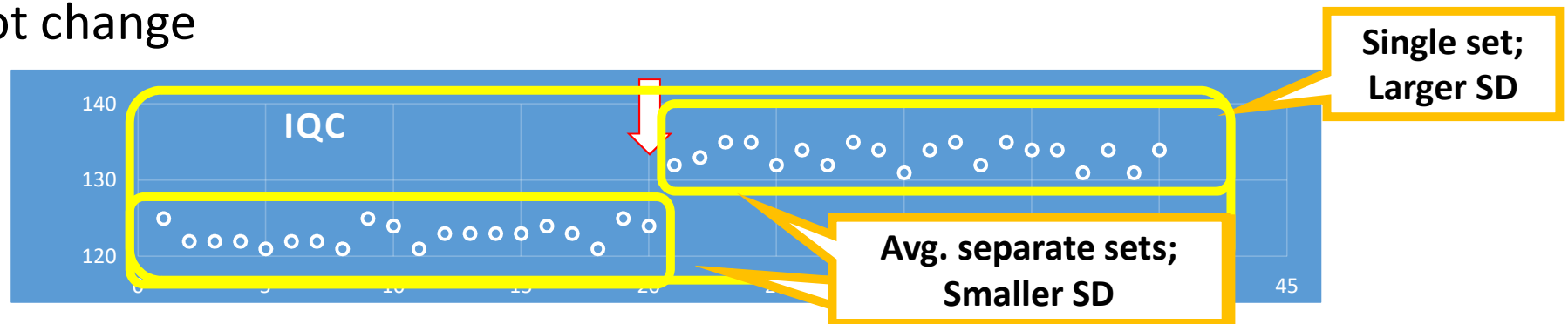
- combination of all uncertainties from the highest available reference (calibrator or measurement procedure) down to the assigned value of the calibrator
- Manufacturer, is responsible for determining u_{cal}
- If laboratory prepares the calibrator, the laboratory is responsible for determining u_{cal}
- For some calibrators a comprehensive u_{cal} is technically unavailable; e.g. some international conventional biological reference materials

Uncertainty of bias, u_{bias}

- IVD manufacturers are responsible for ensuring minimal, medically acceptable measurement bias
- **If ongoing EQA indicates the introduction of a medically significant Bias**

NOTE: Changes during IQC data collection

- Calibrator lot change/Recalibration
- Maintenance/Part replacement
- Reagent lot change
- IQC lot change



NOTE: Changes during IQC data collection

➤ Are **IQC results** and **patient sample results** **affected the same way?**

- Calibrator lot change/Recalibration → **YES.** Usually affects IQC as patient samples. Treat IQC data as a **single set**
- Maintenance/Part replacement → **YES/NO.** Both IQC and patient samples affected: treat IQC data as a **single set**; Only IQC affected: treat IQC data as **separate sets**
- Reagent lot change → **YES/NO.** Both IQC and patient samples affected: treat IQC data as a **single set**; Only IQC affected: treat IQC data as **separate sets**
- IQC lot change → **NO.** Treat IQC data as **separate sets**

